

# Is prediction nothing more than multi-scale pattern completion of the future?

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## ABSTRACT

While the notion of the brain as a prediction machine has been extremely influential and productive in cognitive science, there are competing accounts of how best to model and understand the predictive capabilities of brains. One prominent framework is of a “Bayesian brain” that explicitly generates predictions and uses resultant errors to guide adaptation. We suggest that the prediction-generation component of this framework may involve little more than a pattern completion process. We first describe pattern completion in the domain of visual perception, highlighting its temporal extension, and show how this can entail a form of prediction in time. Next, we describe the forward momentum of entrained dynamical systems as a model for the emergence of predictive processing in non-predictive systems. Then, we apply this reasoning to the domain of language, where explicitly predictive models are perhaps most popular. Here, we demonstrate how a connectionist model, TRACE, exhibits hallmarks of predictive processing without any representations of predictions or errors. Finally, we present a novel neural network model, inspired by reservoir computing models, that is entirely unsupervised and memoryless, but nonetheless exhibits prediction-like behavior in its pursuit of homeostasis. These explorations demonstrate that brain-like systems can get prediction “for free,” without the need to posit formal logical representations with Bayesian probabilities or an inference machine that holds them in working memory.

## 1. Introduction

Several fields in the cognitive and neural sciences have recently been promoting the idea that prediction of sensory input is a fundamental aspect of many cognitive and perceptual processes (Clark, 2015; Friston, 2019; see also Mumford, 1992, and Rao and Ballard, 1999). For example, work in real-time language comprehension suggests that readers and listeners tend to develop anticipations for certain categories of upcoming words (e.g., Altmann and Mirković, 2009; Onnis et al., 2008; see also van Petten and Luka, 2012). Three decades ago, Jeff Elman (1990), (1991) demonstrated that when an artificial neural network uses prediction (instead of pattern classification) as the goal of its output layer, learning of temporal contingencies naturally emerges. In fact, Radford et al.’s (2019) recent prediction-based language learning model, GPT-2, predicts the next word in 720 Wikipedia articles with impressive accuracy (e.g., very low surprisal values). It also mimics human data of neural activation patterns quite accurately for words that are less surprising or more surprising (Schrimpf et al., 2020 bioRxiv). Schrimpf et al.’s comparison of various language models suggests that,

since predictive models (such as GPT-2) fit neural data better than non-predictive models, prediction may indeed be a fundamental mechanism of language processing and language learning (see also: Goldstein et al., 2021, bioRxiv).

In the case of language learning, prediction allows the learner to essentially bypass the *negative evidence* problem. In language learnability theory, negative evidence consists of examples of linguistic constructions that *do not* belong to the grammar and are labeled as such. Traditional nativist accounts of language learning once suggested that, since children appear to receive very few explicit corrections of their ungrammatical utterances (Marcus, 1993), this absence of negative evidence should make it impossible to acquire the language via learning from examples (Gold, 1967). Thus, the language acquisition process was claimed to be largely innate (Pinker, 1994). Nativist linguists used this absence of explicit negative evidence (e.g., absence of a teacher explicitly providing an error signal) as an indication that language is not actually learned but that it simply grows in the body much like the kidney or the liver “in accordance with the genetic program” (Chomsky, 1980). In fact, Chomsky went so far as to suggest, “It is possible that the

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notion ‘learning’ may go the way of the rising and setting of the sun.” (p.139).

However, Elman’s (1990, 1991) prediction-based simple recurrent network serves as an existence proof that some degree of grammatical structure can be learned without explicit negative evidence and without innate grammatical knowledge. Internal negative evidence can be generated by a passive listener who is essentially eavesdropping on a native speaker of the language and generating predictions (or at least implicit anticipations) of possible upcoming sentence constituents. In fact, Spivey-Knowlton and Saffran (1995) described it as essentially getting implicit negative evidence for free, when the network’s output is a prediction of the next input. While the network eavesdrops on a speaker, every little incremental anticipation of a class of nouns next or a class of verbs next gets implicit negative evidence simply due to the fact that the speaker continues the utterance and produces either a verb next or noun next, or something else. The error signal is thus derived not by a teacher, per se, but by the learner simply comparing its various anticipated next inputs to the actual specific next input. These implicit anticipations may be quite distributed and multifarious, a bit like syntactic priming (Pickering and Branigan, 1999) or even recurrent pattern completion (Tang et al., 2018).

If prediction allows the learner to essentially “get negative evidence for free,” is there a way that a learner could, in turn, “get prediction for free” in advance of that? Many of the mainstream treatments of predictive processing in perception and cognition are inspired by *forward models* of sensorimotor action where the network architecture explicitly dedicates a module for the purpose of predicting future sensory events that may result from motor output (Grush, 2004; Jordan and Rumelhart, 1992; Kawato, 1999; Wolpert and Flanagan, 2001). In the context of higher cognition, that prediction module is sometimes treated as generating formal representations of expected sensory inputs (Gładziejewski, 2016; but cf. Kirchhoff and Robertson, 2018). Even researchers who are aimed at neurally plausible accounts of language processing will sometimes hint at explicit representational predictions of future linguistic input by referring to “predictive properties of the higher order context” (Marslen-Wilson, 1973) or “discourse-driven structural expectations” (Spivey-Knowlton, 1992) or suggesting that “by forming explicit predictions of upcoming material, the parser may not only ease the processing of current material (as described earlier), but get a head start on future material” (Kutas et al., 2011). However, it might be possible that a network with some simple pattern completion abilities as part of its architecture, perhaps assisted by recurrent feedback projections (e.g. Tang et al., 2018), could essentially *get prediction for free* – without the need for a module that is dedicated to explicitly generating those predictions. For example, in the case of language comprehension, predictive processing may emerge as a kind of “forward momentum” inherent in the state-space trajectories that comprise the transitions from one word-recognition event to the next in a sentence (Elman, 2004; Elman, 2009; Spivey, 2008). Where exactly would that forward momentum come from? One possibility is neuronal pattern completion processes.

## 2. Pattern completion as a form of prediction

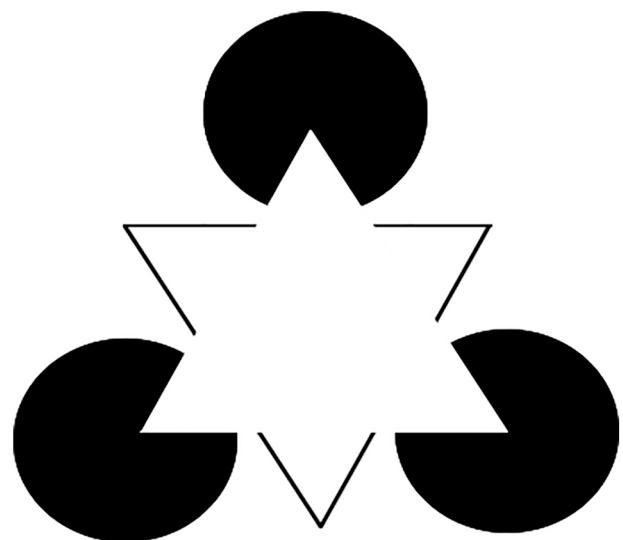
While it may be tempting to conceive of predictive processing as involving an explicit representation of possible future input that is held in working memory and then cognitively compared to the actual future input to compute an error signal (Gładziejewski, 2016), one need not treat predictive processing in quite such an explicitly representationalist fashion (Kirchhoff and Robertson, 2018). For example, Barsalou (2009) offers an account of conceptual representations being contextualized by different situational circumstances, and even generating a form of predictive processing, by relying on an inference mechanism that uses pattern completion to activate a perceptual simulation of the part of a multi-step event that has yet to be perceived: hence, a prediction. While Barsalou places this pattern completion process inside an inference

mechanism, it is possible that the pattern completion process could be intrinsic to the perceptual simulation itself – without the need for a mechanism dedicated to making inferences. In this article, we explore the pattern completion processes that may be intrinsic to the spreading of activation across a population of neurons that respond to a perceptual event, or that might encode a perceptual simulation of that event. We start with pattern completion in visual perception.

Tang et al. (2018) recorded intracranial electric field potentials from the left fusiform gyrus in awake behaving humans while they viewed substantially-occluded pictures of faces. A peak in electric potential from that cortical region was observed 160 ms after stimulus onset when the face was not occluded, and this same peak was observed at progressively longer latencies as the faces were presented with greater amounts of occlusion. Even with only 15% of the stimulus visible, participants were often still capable of recognizing the image as a face, but the peak in electric potential was shifted to around 250 ms after stimulus onset – about 90 ms later. In three feed-forward models of visual object recognition, Tang et al. found that many of these occluded images were not accurately recognized. However, when recurrent feedback connections were introduced into one of those models, its recognition improved significantly and even exhibited a delay in recognition of occluded faces that matched the human data.

Although this pattern completion process entails additional time to achieve recognition (perhaps an additional 90 ms), this additional time is not treated as a prediction of the future but instead simply as a temporally drawn-out process of “filling-in” the gaps in the original visual stimulus. The perceptual effects of filling-in those gaps that are present in a retinal image have often been treated as the result of a pattern completion process whereby the neurons that form the population code for a particular stimulus (e.g., a distributed cell ensemble) pass recurrent activation back and forth among each other to eventually activate even those neurons that were not initially responding because they code for aspects of the stimulus that are currently absent (or occluded) in the visual input (Grossberg, 2003; Mendola, 2003). This filling-in process is strikingly apparent in the Kanizsa Triangle illusion (Fig. 1).

The Kanizsa Triangle illusion commonly induces the perception of illusory contours along the three straight segments of the bright white upright triangle in the center of the image. These illusory contours are proposed to be the result of a pattern completion process whereby a



**Fig. 1.** In the Kanizsa Triangle, a bright white illusory upright triangle appears to be superimposed on top of the upside-down triangle. Despite the absence of a contour along the central portions of its three triangle segments, a faint illusory contour is often perceived there, as if the inside portion of the triangle’s illusory segment is slightly brighter than the outside portion of it.

population code in object recognition areas of visual cortex (such as inferotemporal cortex) becomes active for recognizing the general upright triangle shape and then sends feedback signals that activate orientation-selective cells in earlier regions of visual cortex. Single-cell recordings from the visual cortex of Macaque monkeys have found that cells in area V2 respond to these illusory contours (Von der Heydt et al., 1984), and about half of the cells in area V1 do as well (Grosf et al., 1993) – just with a slight delay in the neighborhood of about 90 ms (Lee and Nguyen, 2001).

Fig. 2 provides an idealized picture of what this pattern completion process might look like over time among the cells of a population code in inferotemporal cortex that recognizes upright triangles. (The cells in this idealized sketch would likely be in inferotemporal cortex, where visual receptive fields are not particularly location-specific, and thus would not actually need to be spatially arranged in the topographical pattern as depicted.) At Time 1, only the nodes responsive to the triangle vertices (present in the “Pacman” shapes of Fig. 1) are active. This is because the nodes responsive to the triangle segments are receiving no input from orientation selective cells in V1 and V2, since there is no actual contour there in the visual input. Soon thereafter, at Time 2, the vertex-responsive cells will have spread their activation to the segment-responsive cells to make them partially active. Thus, the population code for recognizing this upright triangle will relatively quickly pattern-complete itself. After this slight delay, feedback projections from those partially-active segment-responsive cells will send signals back to

orientation-selective cells in V2 and V1, causing them to become partially-active as well – even though their external input does not contain their preferred contour. The information being developed among the cells in Fig. 2 serves as a generative model (Teufel and Fletcher, 2020) that induces a kind of “controlled hallucination” (Clark, 2015) of the visual features needed to complete the picture. In this way, pattern completion causes the neural system to activate stimulus features that are *not actually* present in the stimulus. If the stimulus itself was an event that takes place over time, then perhaps this pattern completion could cause the neural system to activate stimulus features that are *not yet* present in the stimulus.

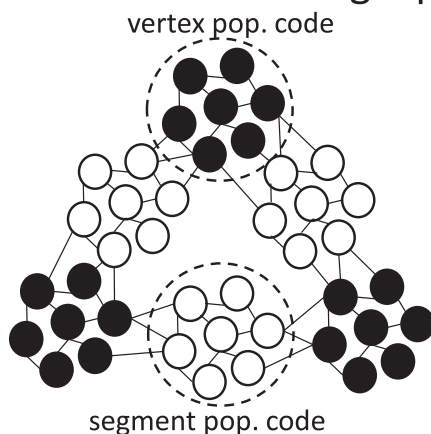
Although predictive coding in these generative models is now the norm in vision research, it is rarely aimed at actually forecasting future stimuli. In fact, Corlett et al. (2019) clearly state that predictive coding means the perceptual system is “adopting a hypothesis that explains what is causing our *current* sensation” (p. 115, italics ours). Visual processes in general, and especially visual illusions that rely on pattern completion, have a tendency to be treated as though they involve a parallel input of all visual information at once. Therefore, when pattern completion takes place in this context, it is rarely treated as a temporally organized process. That is, filling in the missing piece of an image, such as an illusory contour, is not treated as a prediction of something in the future. Nonetheless, eye movements do convert that seemingly parallel input of visual features into a sequential delivery of foveated visual features. The sequence of fixations generates a temporal organization of the visual input that can significantly influence perception and cognition. For example, changing one’s pattern of eye movements can change how the Necker cube is perceived (Pomplun et al., 1996) or change how a diagrammatic problem is solved (Grant and Spivey, 2003) or even reveal the role of pattern completion processes in false alarms during a memory task (Wynn et al., 2020). When some temporal organization is introduced to the process of pattern completion, some of the feature detectors that get activated by their neighboring cells (but not by external input) can be seen as preparing the brain for stimulus features that are yet to come. For instance, in his peer commentary for the Pessoa et al. (1998) treatment of perceptual completion in visual perception, Horace Barlow (1998) wrote succinctly, “Prediction is pattern completion in the temporal domain.”

In contrast to the field of vision research, the field of language research typically uses the term “predictive processing” to explicitly refer to the forecasting of future stimuli. For example, Kuperberg and Jaeger (2016) openly state that “it is logically impossible to explain these [predictive processing] effects without assuming that the context influences the *state* of the language processing system *before* the bottom-up input is observed” (p.33, italics theirs). Kuperberg and Jaeger provide a detailed review of the many different ways in which the term “prediction” has been used in language research. Importantly, they point out that an approach that uses Bayesian belief updating to predictively compute multiple potential future stimuli in parallel, each with graded degrees of belief, can fit a wide variety of human data in psycholinguistics. As a mathematical description of the pre-activation of distributed representations of potential future stimuli, that approach may be consistent with our discussion of neural population codes here.

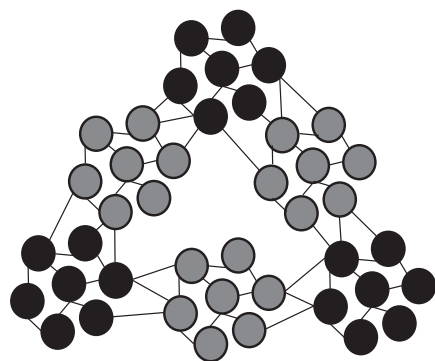
Imagine a population of interconnected neurons that functions as a distributed representation for a common sequence of three events, such as the image of a ball leaving a hand, then a ball in the air, then a ball landing in a different hand – or, alternatively, imagine a Subject Noun followed by a Transitive Verb followed by an Object Noun. There might be a neuronal population code dedicated to recognizing each of those individual sub-events. However, the frequent co-occurrence of the triptych makes for one whole event that is likely to have those individual population codes conjoin to form one larger population hypercode that functions as a distributed representation of that basic throw-and-catch event – or that basic NVN sentence structure (see Elman and McRae, 2019).

In Fig. 3, we provide a schematic depiction of a transition of

## Time 1: Kanisza triangle presented

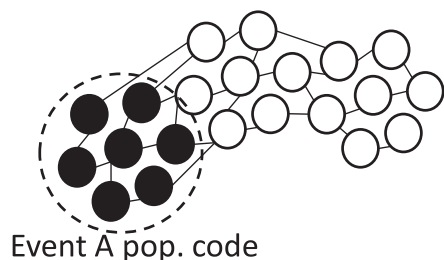


## Time 2: Pattern completion



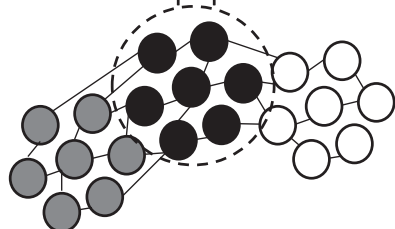
**Fig. 2.** A stylized depiction of a population code for the illusory white triangle in the Kanisza diagram. Nodes in black are highly active due to external input matching their receptive field selectivity. Soon after, activation spreads from those vertex-responsive cells to the segment-responsive cells (in grey) via the process of pattern completion. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

## Time 1: Event A presented

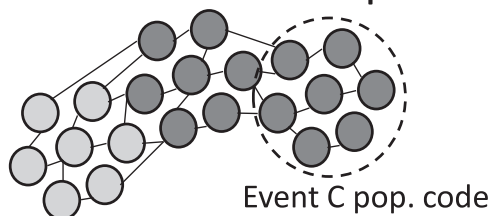


## Time 2: Event B presented

Event B pop. code



## Time 3: Event C not presented



**Fig. 3.** A stylized depiction of a population hypercode for the sequence of Events A-B-C (such as a Noun-Verb-Noun sentence). Nodes in black are highly active due to external input. At Time 3, with no external input, population codes for Events A and B exhibit residual activation (in grey), and population code C exhibits anticipatory partial activation due to pattern completion. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

activation patterns across a population hypercode – or what [O'Reilly and Rudy \(2001\)](#) call a “conjunctive representation.” This could be a set of population codes that respond to the class of Subject Nouns, the class of Transitive Verbs, and the class of Object Nouns. If only the first and second sub-events are presented to the sensory system (e.g., a Subject Noun and then a Transitive Verb), pattern completion will cause this population hypercode to naturally begin activating the portion of itself that represents the third sub-event (e.g., the class of Object Nouns). In this way, pattern completion causes the system to activate event features that are not yet present in the stimulus. One may be tempted to call this anticipatory activation a “prediction,” but it need not be reified as an explicit representation of expected future input that is intentionally placed in working memory, or inside an inference mechanism, or anywhere on the stage of a Cartesian Theatre in the mind. In the spreading of activation across this population hypercode, this anticipatory activation is nothing more than the inevitable result of the common perceptual process of pattern completion.

What allows this population hypercode to have the right synaptic connectivity for achieving this apparent predictive processing is, of course, the organism's adaptation to its environment, i.e., all the learning that has wired its brain to know that subject nouns tend to be

followed by verbs which tend to be followed by object nouns. As the organism gets entrained by its environment, the state space trajectory of its perceptual/cognitive processing “leans forward in time” with a kind of momentum that, to an outside observer, resembles prediction. But inside the neural network of the organism, it need not be anything more than a distributed spreading of activation carrying out a form of “linguistic pattern completion” (Ferreira & Qiu, submitted; [Huettig et al., submitted](#)). In fact, the apparent predictive processing need not involve any formal symbolic representations of the “predicted” sensory input at all ([Kirchhoff and Robertson, 2018](#)).

### 3. Entrainment between two subsystems can generate “Predictive” processing

One area where the vision research community focuses on temporal organization in pattern completion is the work on representational momentum ([Freyd and Finke, 1984](#); [Jordan and Hunsinger, 2008](#)). In visual perception, representational momentum is the tendency to judge or remember the position of an object as further along in the path of the object's movement than was actually presented. The primate visual system has many interconnected motion-sensitive cells with direction-selective receptive fields, so when a moving stimulus suddenly vanishes, the visual system cannot help but continue its perceptual simulation of the motion slightly further along than what actually happened. Evidently, pattern completion not only “fills in” missing information, but can also “forward fill” information, anticipating upcoming, not-yet-presented stimulation ([Spivey, 2008](#), chapter 8). These completion processes are dynamically assembled, possibly with sensitivity to global structure. Considering this pattern completion process as predictive, as we do, is a highly generalizable perspective, extendable to anticipatory momentum and pattern completion in higher-dimensional domains such as conversational turn-taking or collaborative problem solving.

There is a plethora of theoretical and mathematically explicated examples supporting predictive pattern completion, particularly in loosely coupled systems. Generally, a loosely coupled system is composed of a “leader” subsystem that is connected to a “follower” subsystem. The leader's states are independently driven, whereas the follower's states are determined by the leader's states to some degree. (In the context of an organism in a large complex environment, one could loosely treat the organism as the follower who must, at times, anticipate events in the environment, or leader.) Under certain coupling relations, the follower subsystem will become entrained to the rhythms of the leader in a fashion that allows the follower to exhibit anticipation of the leader's states. [Stepp and Turvey \(2010\)](#) present some paradigmatic examples of coupling relations in loosely coupled systems that can yield anticipatory pattern completion. One example involves two external-cavity diode lasers with chaotic (and therefore unpredictable) regimes. The follower laser receives input about the leader laser's present state with negligible delay, and thereby the lasers' activities are synchronized and identical. When the leader laser received feedback at some delay, however, the follower laser began to show a negative phase shift. In other words, while receiving the same input, now from a leader system that receives feedback, the follower laser was able to anticipate the leader laser's future states. A second [Stepp and Turvey \(2010\)](#) example involves Rössler systems, a simple system of ordinary differential equations which exhibits chaotic dynamics ([Park, 2005](#)). This time, feedback about the leader subsystem was embedded into the follower subsystem. Again, the follower exhibited anticipation of the leader's future states – even though the leader subsystem was producing chaotic behavior. Thus, lasers and ordinary differential equations, using simple feedback mechanisms, are able to exhibit a form of anticipatory processing – obviously without utilizing any “mental representations” of predictive coding.

These examples fit well into [Dubois's \(2003\)](#) definition of systems that engage in “strong anticipation.” These are systems where future states are not estimated, modeled, or represented by explicit prediction,



rather, future states are directly anticipated by feedback and the system's constraints. Strong anticipation differs from what Dubois calls "weak anticipation," a system with an explicit internal representational model of the sequential patterns in the environment. By contrast, a strongly anticipatory system's ability to show emergent anticipatory behavior is dependent on its coupling relationship with the system it is "predicting" (Grover et al., 2018). For example, as a neural network adapts to its environmental inputs, it becomes loosely coupled with that environment. It may anticipate a future environmental input simply as a result of its entrained synaptic connectivity. This coupling and entrainment allow pattern completion to spread activation to feature detectors that are attuned to likely upcoming stimuli. The concept of coupling relationships is extremely flexible, readily applicable to a wide variety of contexts for describing emergent predictive processing via a generalized form of pattern completion.

In biological systems, where not every subsystem is fully explicable, coupling relationships can still serve as a lens for examining emergent predictive processing resulting from pattern completion. In an experiment partly inspired by the work of Dubois (2003), Stephen et al. (2008) asked human participants to tap along to a chaotic auditory metronome (its beat onsets followed the chaotic Ikeda map). The participant was the follower system and the chaotic metronome was the leader system. The participants' tapping was driven by the metronome, and the participants' ability and effort in following the metronome constituted their coupling relation. Stephen et al. applied detrended fluctuation analysis (DFA) to the tapping and to the metronome signals. DFA carves up a time series into many small time-windows, and then again in slightly larger time-windows, and again in progressively larger time-windows. With each of these fine-grain and coarse-grain binned time series, DFA computes the root mean squared error (RMSE) of a linear fit to the binned data. When those various RMSEs are plotted on log-log coordinates, the slope of that line is a scaling parameter that indexes long-range correlations in the original time series (e.g., repeated patterns at long time scales). In this analysis, Stephen et al. found a high correlation between the metronome's scaling parameter and each participant's tapping scaling parameter. Thus, even though the precise timing of each beat from the chaotic metronome was itself unpredictable, participants were nonetheless attuning their tapping to the overall statistical structure inherent in their input (see also Rigoli et al., 2014; Schloesser et al., 2019). In a similar analysis, Marmelat and Delignières (2012) examined interpersonal synchronization between two humans as they swung pendulums in synchrony (where neither agent is the leader or follower). They found a similar long-range statistical attunement process emerging between the two participants in a dyad, suggesting that their rhythms were becoming entrained with one another and they were beginning to function as one system (Spivey, 2020).

In addition to shared pendulums, interpersonal interaction has also been studied with natural conversation among dyads. For example, a listener's time series gets coordinated with a speaker's time series in their postural sway (Shockey et al., 2003), their eye movements (Richardson and Dale, 2005), their facial and hand gestures (Louwerse et al., 2012), and even their brain activity (Kuhlen et al., 2012), with a relatively short lag between speaker and listener. If the conversation is mutual and bidirectional, that lag disappears and the interlocutors are synchronized with essentially zero lag (Richardson et al., 2007). When those two conversational partners are not functioning as a leader system (speaker) and a follower system (listener), but instead as mutual co-contributors to the dialogue, then zero-lag synchrony (e.g., of eye movements, gestures, spoken words) would be impossible without some form of linguistic anticipation (Emberson et al., 2010). The question we wrestle with here is the following: How much of that anticipation is due to formal internal representations that model the sequence of events and make explicit predictions (Gładziejewski, 2016), and how much of that anticipation is a form of *strong anticipation* (Dubois, 2003) in that it is simply due to the organism attuning itself to the temporal statistics of the environment and engaging in pattern completion over time?

#### 4. Pattern completion through time in language

With prediction in visual perception, lasers, and even ordinary differential equations all exhibiting a kind of pattern completion through time in a fashion that need not involve anything resembling conscious cognitive processing, is it possible that language, too, can have its predictive processing explained in a fashion that requires little in the way of formal logical representations and Bayesian probabilities that are held in working memory? To begin to see how this might be, consider what happens when a speaker gets through most of a sentence but suddenly finds herself failing to find the right word for the end of the sentence, and then a well-entrained listener completes that sentence with the correct missing word. As discussed in the previous section, if a leader subsystem were to suddenly freeze and produce no behavior for a short period of time, the follower subsystem will nonetheless produce some behaviors that would later be seen as very similar to those of the leader once it resumed its activity. Thus, the follower would essentially be predicting the leader's behavior during that interregnum. The notion of "strong anticipation" (Dubois, 2003) could be naturally extended to explain evidence that language processing involves a prediction of upcoming input.

Predictive models of language, and of cognition more generally, have had a great deal of success in modeling human behavior, and thus are worth examining more closely, before we suggest some refinements in their interpretation. Consider again Elman's (1990, 1991) simple recurrent network (SRN), one of the seminal language models in the field of connectionism. This network is "simple" in that it contains just three layers: an input layer, a hidden layer, and an output layer. One key innovation of this model was the inclusion of a set of "context" nodes in the input layer, which are just a copy of the hidden layer's activity from the *previous* timestep. This allows the model to use past information to influence processing in the present. The other key innovation was the use of each *subsequent* input as the training signal: in the SRN, the most recent output is compared with the next input, generating an error signal that guides learning. In other words, each output is treated as a prediction of the next input, and the model learns by virtue of minimizing prediction errors. With these simple mechanisms, Elman's SRN built up hidden-layer representations that resembled knowledge of grammatical classes, despite having no information about semantic meanings (but cf. Howell et al., 2005; Huette and Anderson, 2012; and Li and Joanisse, 2021, for recurrent networks that develop semantic relationships). The network also offered an account of several phenomena observed in human language processing, such as higher prediction errors near the onsets of words and sentences relative to the middle or end of them, which is consistent with results of eyetracking and neuroimaging studies (Smith and Levy, 2013; Staub and Clifton, 2006; Van Berkum et al., 2005).

More recent deep-learning neural networks that also train on predictions, such as BERT and GPT-3, show astoundingly human-like capacities for language production and "comprehension" (e.g., generating summaries of, or responses to, text input), and the activation patterns they exhibit closely match human neural responses to the same stimuli (Schrimpf et al., 2020). Results of this kind have led to widespread acceptance of the idea that prediction of upcoming inputs is a crucial aspect of the human capacity for language (for some examples, see: Ryskin et al., 2020a; Ryskin et al., 2020b; Kamide et al., 2003; Kuperberg and Jaeger, 2016; Staub and Clifton, 2006; for a critical take complementary to ours, see: Huettig and Mani, 2016).

The general class of models that represent cognitive systems as generating explicit predictions of future input and learning from prediction errors can be classified as *predictive processing* models (Elman 1990; see also: Glimcher, 2011). These can be distinguished from a more specific subclass of *predictive coding* models, in which the signals that pass between processing units are themselves prediction errors (Clark, 2015; Luthra et al., 2021a). Predictive coding models have been highly influential across domains of cognitive science, beginning with the work

of Rao and Ballard (1999). Rao and Ballard proposed an inversion of the traditional “bottom up” models of visual processing (Marr, 1980), suggesting that feed-forward connections beginning at the retina do not carry visual information per se, but rather prediction errors: the mismatch between the top-down hypotheses of current visual stimuli and the *actual* current stimulus. On this view, visual perception becomes a sort of “controlled hallucination” (Clark, 2015), where the visual system is constantly generating the images it expects to see, and adjusting those expectations only when they are violated. More recently, the notion of prediction error minimization has been taken up by proponents of the “Free Energy Principle,” which has been heralded by some as a possible “grand unifying framework” for cognitive science, offering accounts of how reduction of sensory prediction error may be responsible for everything from perception-action loops, to psychiatric illness, to consciousness, and to the emergence of life itself (Clark, 2015; Friston, 2010; Wiese and Friston, 2020; Ramstead et al., 2018; Veissière et al., 2020).

Since the time of Elman’s (1990, 1991) and Rao and Ballard’s (1999) work, prediction error minimization has become one of the most popular frameworks for understanding language production and comprehension. One prominent example is the Rational Speech Act (RSA) framework (Frank and Goodman, 2012), which offers a model of pragmatics in the tradition of Grice (1989). In the RSA framework, pragmatic speakers attempt to communicate something about the state of the world by selecting an utterance that would minimize prediction error (here called “surprisal”) for the listener, while accounting for the effort required of the speaker. To do so, speakers “simulate” generating each possible utterance, calculate the likelihood of each state of the world being referred to with that utterance, and subtract the cost of the utterance. This gives a measure of the utility of each utterance, and speakers then select an utterance with probability proportional to its utility. In experimental settings with a restricted set of options for utterances and interpretations, this model has been highly successful at predicting human behavior where previous models have failed, such as in the case of non-literal or “polite” (i.e., indirect) language (Yoon et al., 2020).

Clearly, explicit representations of probabilities are a crucial aspect of Bayesian models such as RSA, placing them into the category of “weakly anticipatory” systems (Stepp and Turvey, 2010). Whether human brains actually implement something like Bayesian inference or whether this is merely a useful abstraction has been the subject of much debate (Andrews, 2020; Bruineberg et al., 2020; Jones and Love, 2011; Griffiths et al., 2010; McClelland et al., 2010). One prominent challenge to the Bayesian view is a philosophical one: The notion of explicit representations of probabilities that are delivered to a central executive seems, to some, to imply a sort of dualism—a Cartesian “theater of the mind”—that could be taken as an instance of the “homunculus fallacy” (e.g., Monsell and Driver, 2000).

Other challenges are empirical: Obviously human behavior is not perfectly rational, and moreover, the computational requirements of Bayesian inference grow exponentially with the number of variables involved, quickly exceeding the computational capacity of *any* physical system for even simple problems (Sanborn and Chater, 2016). Sanborn and Chater have argued that many of these critiques can be overcome if we view brains as Bayesian *samplers*, which may not need to explicitly represent or calculate probabilities. A Bayesian sampler can make inference computationally tractable by “stepping” only locally around a probability distribution in search of peaks or troughs, and in the process can reproduce many systematic reasoning errors observed in humans. Many of the human-like errors of the sampling algorithms relate to the way that subsequent samples are autocorrelated, which Sanborn and Chater associate with priming phenomena in cognitive psychology. However, it’s not clear if much is gained by calling brains Bayesian samplers, given that the priming-like phenomena (i.e. autocorrelated observations) that Sanborn and Chater discuss already fall so naturally out of connectionist systems (Rueckl, 2003; Stark and McClelland, 2000). Thus, if Bayesian sampling were occurring in brains, a plausible

mechanism for it could be pattern completion as we’ve described it: each “sample” drawn may simply be a new pattern cohering – an attractor basin being reached – as the brain moves through its activity state-space, directed partly by its own anticipatory momentum.

Once one has accepted the role of pattern completion through time in visual perception, it is not much of a stretch to see how this account can extend to other domains, including language. Indeed, pattern completion has been proposed to take place not only in neural mechanisms of visual perception but also in the neural mechanisms of memory and other perceptual/cognitive processes (Marr et al., 1991; O’Reilly and McClelland, 1994). Auditory pitch perception offers one clear example (Hubbard, 1996; Johnston and Jones, 2006). Jones et al. (2006) found that when people hear a sequence of tones that either ascends or descends in pitch, they tend to judge a new tone as higher or lower (relative to its actual pitch), respectively. In another experiment, when participants heard a sequence of tones that regularly and periodically ascended and descended, they judged a new tone in accordance with the global structure and pattern (Johnston and Jones, 2006). The sensitivity to pattern completion persisted even when their judgement violated the preceding pitch momentum and direction, indicating that judgments were influenced by global and not just local patterning.

These results from auditory pitch perception are analogous to findings in the psycholinguistic literature on sentence processing and, at a shorter time-scale, in the processing of individual words. Consider the famous N400 ERP component, which has been taken as a measure of expectation violation. For example, when participants hear a sentence such as “I take my coffee with cream and *dog*,” a larger N400 is observed than when the final word is “sugar,” suggesting that participants *expect* to hear “sugar” (Kutas and Hillyard, 1980; Rabovsky et al., 2018; Rabovsky and McRae, 2014). But what if the expectation of an upcoming word begins as a form of automatic priming and perceptual pattern completion, which only later becomes an explicit prediction that one is consciously aware of?

Let us zoom in to the millisecond time scale and examine the processing of an individual spoken word. When one hears only the first couple hundred milliseconds of a word fragment, such as “eleph”, there is substantial psycholinguistic evidence showing that the unspoken remainder of that word gets partially perceptually activated anyway (e.g., Grosjean, 1980; Marslen-Wilson, 1987; Allopenna et al., 1998). Such a result is to be expected if the primary unit of representation for a word is a distributed population code or cell assembly (Pulvermüller, 1999), as opposed to an individual “grandmother” cell for each word (Bowers, 2009). Through Hebbian learning, individual neurons that may be responsive to different aspects of a particular word (e.g. phonology, orthography, syntactic class, meaning, etc.) will strengthen their connections with each other, resulting in a distributed functional assembly that can be taken as a representation of the word (for a compatible computational model, see: Kawamoto, 1993). As a result, partial activation of a subset of such an assembly can spread into a complete pattern over the time course of a few hundred milliseconds.

The type of pattern completion that occurs in Hebbian cell assemblies that support language processing may not appear on face value to be predictive but could nonetheless indirectly result in predictive processing. To see more clearly how this may be the case, let us consider findings obtained with a “gold-standard” model of spoken word recognition, TRACE (McClelland and Elman, 1986). TRACE is a neural network model with three layers of nodes representing acoustic features, phonemes, and words, respectively. TRACE uses localist representations, such that each node has a label indicating what linguistic unit it represents, whether that be a phonetic feature, a phoneme, or a word. Units in the first two layers have feedforward connections to units in the subsequent layer of which they are a part. For example, in the phoneme layer, the phoneme unit representing the /r/ sound will have excitatory connections to the nodes for words like “room” and “broom” in the word layer because the /r/ phoneme is present in these words. There are also feedback connections from the word layer to the phoneme layer, such

that activation of the word “room” will also send activation to the other phonemes besides /r/ that are part of the word. Finally, TRACE also has inhibitory connections between nodes within a single layer, such that activation of the words “room” and “broom” will result in competition between these words, with only one word eventually achieving full activation. There are other models of spoken word recognition that do not rely on feedback from the word layer to the phoneme layer (Norris, 1994; Norris and McQueen, 2008), but they have difficulty accommodating the evidence that supports this feedback (Elman and McClelland, 1988; Luthra et al., 2021b; Magnuson et al., 2003; Gow and Olson, 2016). Critically, unlike Elman’s (1990) SRN, which explicitly uses prediction in its architecture and its learning algorithm, TRACE does not predict. In fact, TRACE does not even learn—the connection weights between units are instead coded by hand.

The localist representations used in TRACE are purely for ease of analysis. The theory of parallel distributed processing (Rumelhart and McClelland, 1986) from which TRACE sprang certainly does not posit individual neurons devoted to individual cognitive representations. When one looks at an individual node in the TRACE network, one should imagine a collection of neurons, i.e., a population code, that would correspond to it in a human brain. Models of language processing based on deep learning have greater realism with respect to the distributed population codes found in the brain but are also more difficult to analyze because the representations cannot be neatly carved up. For example, Magnuson et al (2020) recently created the EARSHOT model, a 2-layer LSTM network which uses a spectrographic representation of actual audio speech data as its input. EARSHOT is able to recognize the words in actual speech, rather than being fed a collection of idealized phonetic feature values on its input layer. Within the model’s single hidden layer of 512 nodes, distributed representations emerge in which word-level, phoneme-level, and phonetic-feature-level information are all intermixed. These representations are certainly more similar to the population codes of the human brain than are the labeled localist nodes inside TRACE. However, EARSHOT’s hidden-layer nodes have no labels, which can make it difficult to analyze in detail the population coding that it employs, or to examine if it exhibits an emergent form of predictive processing. Therefore, for our purposes here of examining how predictive processing may emerge from pattern completion alone, we start with the idealized TRACE model of spoken word recognition.

Simulations from the TRACE model mimic a wide variety of psycholinguistic findings in spoken word recognition, such as a lexical influence on phoneme perception, a tendency towards categorical perception, trade-offs between acoustic cues in phoneme identification, and more (McClelland and Elman, 1986). Critically, for our present purposes, TRACE also exhibits properties of pattern completion. For example, Frauenfelder and Peeters (1998) conducted a simulated version of the gating task, in which a listener is presented with longer and longer segments of a word until the word can be identified (Grosjean, 1980). Just as with human listeners, TRACE required only partial information (up to a “uniqueness point,” where the information given becomes consistent with only a single word) in order to identify the word. More recently, Magnuson (2015) showed that TRACE also mimics the phoneme restoration effect, in which a phoneme in a speech stream is replaced by noise, such as a cough, but is nonetheless perceived to be present based on contextual information (Warren, 1970). These effects are not that different from the visual pattern completion phenomena discussed in Section 2. In all of these cases, partial information leads to a pattern of activity over time that resembles the pattern that would be seen if a complete visual or auditory pattern had been presented.

A key pattern that has been taken to support predictive coding (the subclass of predictive processing models in which prediction errors are the signals between processing units) in the brain is a *reduction* in signal when stimuli are predictable, such as the reduced N400 for semantically regular sentences (Kutas and Federmeier, 2011) or reduced reading times for predictable words (Smith and Levy, 2013). This pattern is hypothesized if, as the prediction-error-minimization framework holds,

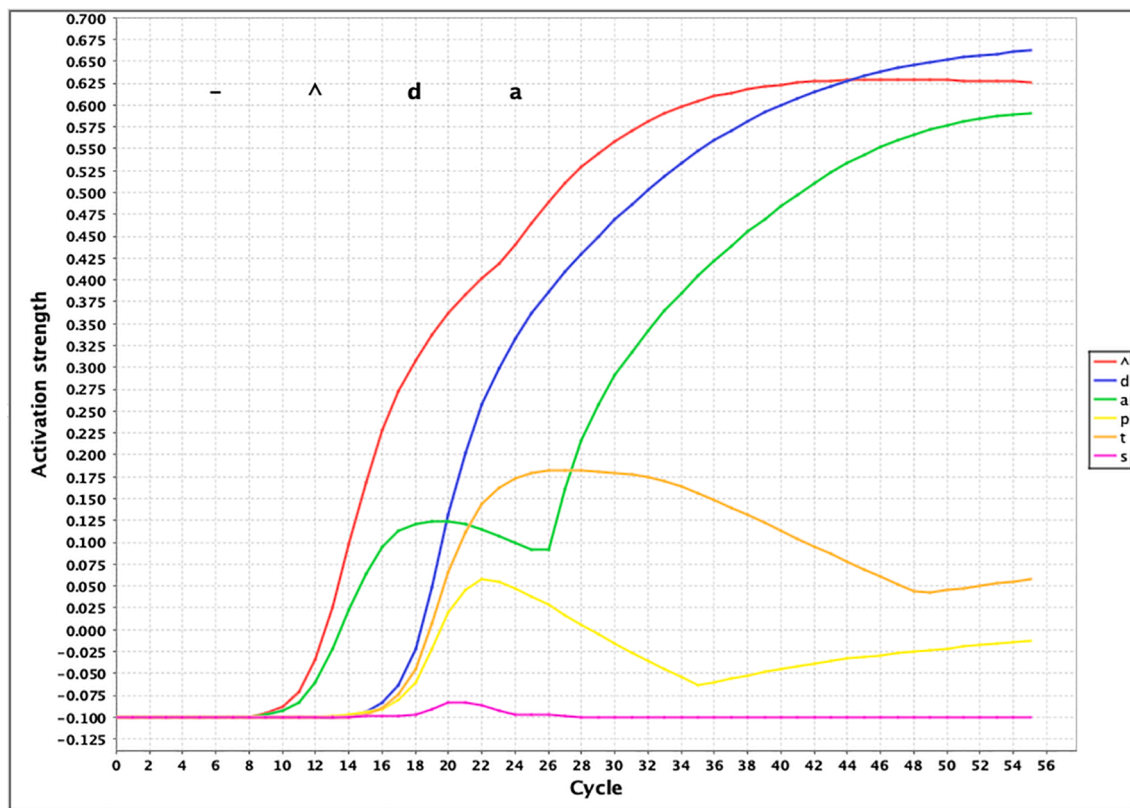
feed-forward connections in the brain primarily carry error signals, which should be smaller in magnitude when expectations are met (Clark, 2013; Rao and Ballard, 1999). Nonetheless, Luthra et al. (2021a) recently reported analogs of this prediction-error signal inside TRACE, which does not make explicit predictions. Luthra et al. found that both the total amount of lateral inhibition at the word layer and the total amount of feedback from the word layer to the phoneme layer, two indices of competition and activity in TRACE, were reduced when the incoming phonetic features were predictable. This occurs because predictable inputs are, by definition, inputs for which the information contained in later segments is partially redundant with respect to early segments. In other words, the guess that a rational agent would make based on partial information is likely to be correct, and full information merely corroborates this guess. So when TRACE gets predictable incoming phonetic featural input, this means that the true lexical target is likely to be among the set of lexical nodes already most strongly activated based on early word segments, and later segments will only confirm a winner, if necessary. But when inputs are less predictable, this means that bottom-up input in later segments is *inconsistent* with the set of lexical candidates that first became active, leading to more lexical nodes becoming active and greater competition occurring among them. Thus, a reduction in signal for predictable inputs (i.e., a signature of predictive coding) can also emerge from a pattern-completion system such as TRACE, without recourse to prediction error signals.

Luthra et al. (2021a) compared the dynamics seen in TRACE with an SRN, which *does* make explicit predictions. However, Luthra et al. point out that the error signal of an SRN is essentially external to the network—it is not a function of the activity inside the network, and does not even need to be present for a trained model to function (in fact, error signals are typically absent during testing). The nodes of the SRN’s hidden layer, which are more analogous to population codes in the brain, do not show the signal reduction that is the hallmark of predictive coding—if anything, quite the opposite. Thus, it appears that the neural signatures of predictive coding in the brain may actually be more consistent with the pattern-completing, non-predictive behavior of TRACE than with the explicitly predictive behavior of an SRN.

Figs. 4 and 5 demonstrate example results of this pattern-completing non-predictive behavior in TRACE that nonetheless allows predictive processing to emerge. Using jTRACE, a version of TRACE implemented in Java (Strauss et al., 2007), we fed the network input corresponding to the acoustic features of the first three phonemes of the word “adapt.” In Fig. 4, the “~” symbol corresponds to the schwa vowel /ə/ at the onset of the word, which is then followed by the phonemes /d/ and /a/. The remaining phonemes /p/ and /t/ were not fed into the network, thus making the input equivalent to someone saying “ada” and stopping before finishing the word *adapt*. The red line in Fig. 4 shows the activation of the /~/ schwa phoneme node rising first. The green line shows the activation of the /a/ phoneme node also rising somewhat at that time due to its shared acoustic features with the /~/ phoneme input. The blue line shows the activation of the /d/ phoneme node rising next. At that time, the phoneme nodes for /p/ and /t/ (yellow and orange lines, respectively) also rise slightly due solely to the fact that they share acoustic features with the phoneme /d/. Then, the green line rises further when the phoneme /a/ is actually input. Since no more external input is delivered, the nodes for phonemes /p/ and /t/ (yellow and orange) reduce in activation for several time steps. However, once the node for “adapt”, in the word layer of TRACE, becomes sufficiently active, it sends feedback to all its constituent phonemes, including those that were not actually present in the stimulus. Thus, we see the activation of the /p/ and /t/ nodes (yellow and orange) beginning to rise again in the later timesteps. Multi-level pattern completion in TRACE has caused it to prime activation of two phoneme nodes that should, in fact, be expected in the upcoming input. This amounts to a form of predictive processing emerging from a network that does not explicitly use prediction.

Similar results are observed in Fig. 5, where acoustic feature input is





**Fig. 4.** TRACE model activations of nodes in the phoneme layer as acoustic feature input is delivered for just the first three phonemes in the word “adapt.” Note how the phoneme nodes for /p/ and /t/ (yellow and orange lines) begin to rise slightly in activation at time steps 35 and 49 even though they were not present in the stimulus. (The phoneme node for /s/, in purple, is shown as a baseline.) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

delivered to TRACE corresponding to only the first four phonemes of the word “progress.” Activation flows up from the acoustic feature layer to the phoneme layer, activating first the phoneme node for /p/ (the red line in Fig. 5). The phoneme nodes for /t/ and /g/ (cyan and yellow lines, respectively) become slightly active at that time as well because they share some acoustic features with the phoneme /p/. Then, when the acoustic input for the phoneme /r/ is delivered, the phoneme node for /r/ (blue line) rises in activation. What rises next is the phoneme node for /a/ (the short vowel pronunciation of the “o” in “progress”, in green). When the final acoustic input is delivered, for the phoneme /g/ (in yellow), its phoneme node rises as well. Several time steps later, however, the phoneme nodes for the schwa phoneme (/ʌ/, in orange) and the /s/ phoneme (in purple) begin rising in activation even though those phonemes were not present in the input.

It is worth noting that the TRACE model generates new copies of its nodes as time progresses, in order to allow it to accurately process words that have repeated phonemes. Thus, when the /p/ and /t/ nodes begin to rise late in Fig. 4, it is actually later copies of those nodes that are receiving the feedback from the word layer and increasing in activation, not the original nodes. Figs. 4 and 5 simply draw one line for a given type of phoneme node, plotting the activation of whichever of them is the most active at a particular time slice. For more detail in the “adapt” simulation, shown in Fig. 6, as the original /p/ and /t/ nodes (in green, which were activated due to their featural overlap with the /d/ phoneme) are reducing in activation, new copies of nodes for the /p/ and /t/ phonemes (in purple) become active around time slice 30 due to feedback from the word layer.

The anticipation of those unrepresented speech tokens emerges due to pattern completion resulting from feedback connections from the word layer to the phoneme layer in TRACE. If one conceives of the mental representation of a spoken word as consisting of a population of

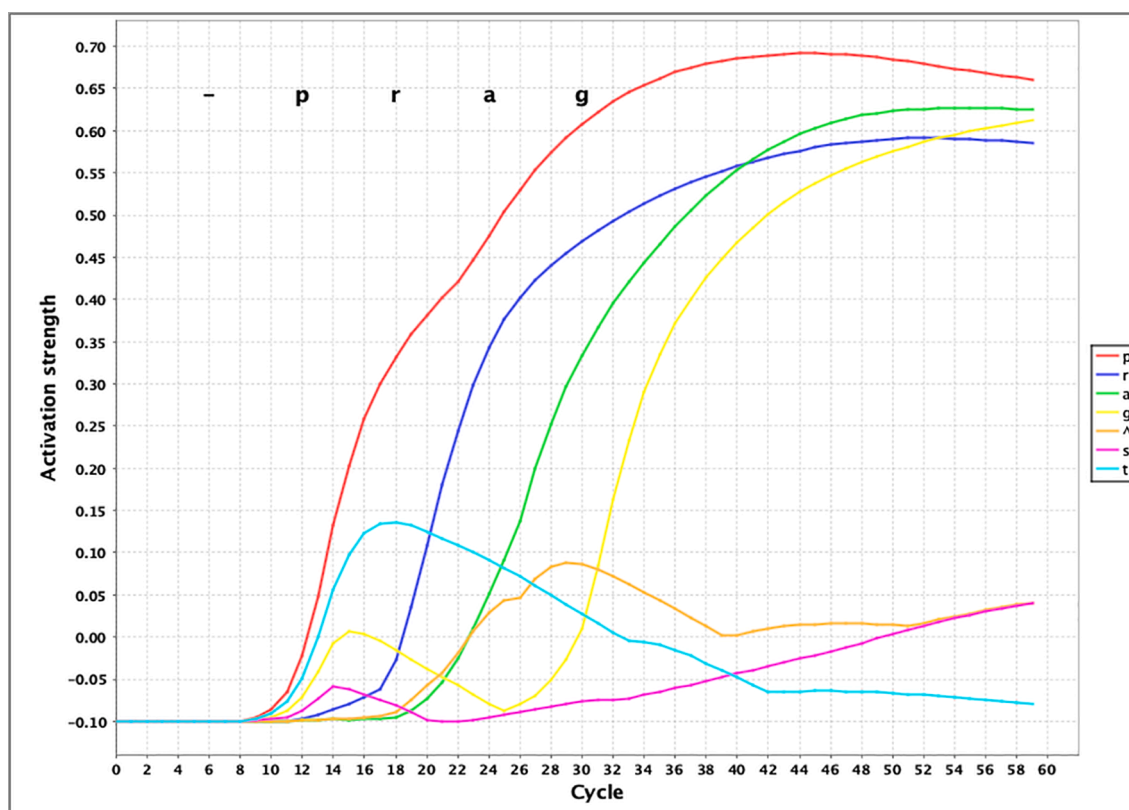
bidirectionally connected neurons that code for speech properties at the scale of acoustic features, at the scale of phonemes, and at the scale of words (e.g., Gow and Olson, 2016; Pulvermüller, 1999), then this population code representation is naturally capable of anticipating, or predicting, future phonemic input simply due to its pattern completion processes. The network exhibits this predictive processing without the need for a mechanism that is dedicated to making predictions. It gets prediction for free.

## 5. Homeostasis is sufficient to generate predictive processing

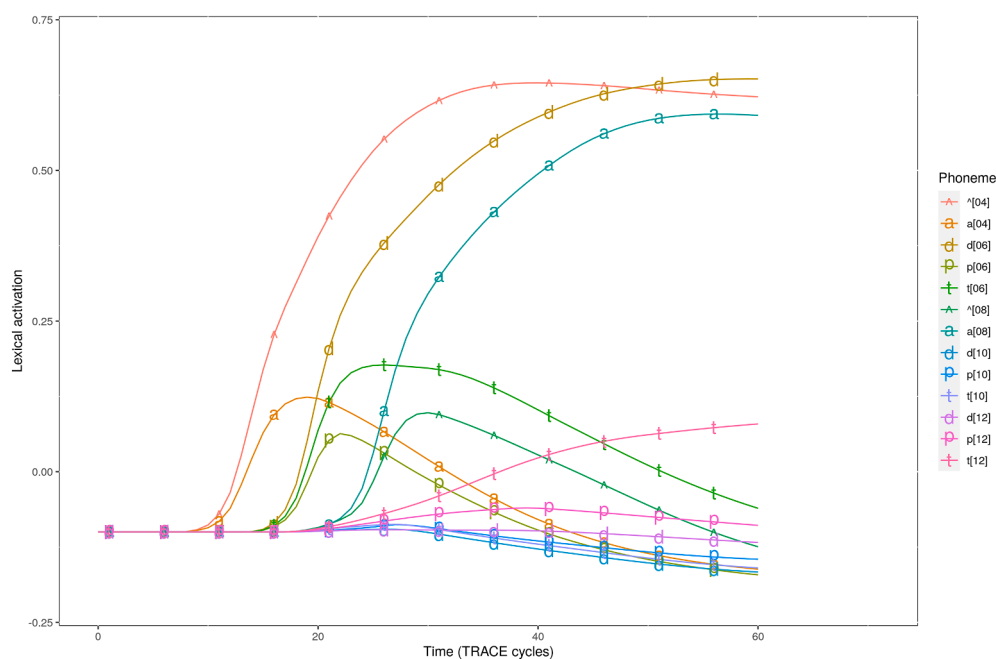
While we have described the spreading activation and competition of TRACE as a form of pattern completion, it is certainly possible to cast a predictive interpretation as well. One could think of the activation of phoneme nodes as evidence about the identities of phonemes and words that haven’t yet been observed, and the dynamics of TRACE as the weighting of different hypotheses. But the point we wish to emphasize here is that this functionally-predictive behavior can emerge from rather “dumb” systems – ones that don’t know they’re predicting. That is, the mechanism for producing behavioral signatures of predictive processing need not be literally a prediction algorithm.

In fact, even the use of labeled, localist nodes as in TRACE is quite a bit smarter than is needed for functionally-predictive behavior to emerge. Consider that neural representations in the brain are much more distributed than those of TRACE, composed of population codes, and none of the individual neurons, nor even the entire brain, needs to know that it is representing anything at all. When we zoom in on the brain, we see living cells, competing for resources and trying to maintain homeostasis in a rapidly changing environment, rather than tiny prediction engines. In this section, we introduce a computational model designed to illustrate how functionally-predictive behavior may emerge in a





**Fig. 5.** TRACE model activations of nodes in the phoneme layer as acoustic feature input is delivered for just the first four phonemes in the word “progress.” Note how the nodes for the schwa /*ʌ*/ and the /*s*/ phonemes (orange and purple lines) begin to rise slightly in activation around time step 40 even though they were not present in the stimulus. (The phoneme node for /*t*/, in cyan, rises initially due to its acoustic overlap with /*p*/ but gradually dies down as it is neither perceived nor anticipated by the network.) (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)



**Fig. 6.** For the “adapt” simulation, temporal copies of phoneme nodes are color coded here to show how it is later copies of the /*p*/ and /*t*/ nodes (in purple) that become active due to feedback when the input is /*ʌ*/ followed by /*d*/ followed by /*a*/ (i.e., the first three phonemes of the word “adapt”). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

distributed network of simple, homeostatic systems, like neurons. However, this model also serves to illustrate that the functional-level description may only hold under certain conditions, and requires an outside observer to interpret it as a prediction. In other words, while we can *read* the model's behavior as prediction, there are no predictions represented inside the system.

Neurons are often studied as homeostatic or allostatic systems (MacLean et al., 2003; Turrigiano and Nelson, 2004; O'Leary and Wylie, 2011): all neurons need to remain within some viable range of activity or else atrophy and die, but the "preferred" range of activity is highly variable across individual neurons. Neurons also adapt to perturbations so as to maintain viability. That is, they have self-preserving mechanisms (e.g., modifying synaptic receptors, changing membrane potentials, etc.) that allow them to retain viability under variable conditions. Adaptation of this kind can be thought of as a form of implicit memory, because it means that changes in response to past inputs are carried into the future and will influence future outcomes. Successfully adapted individuals may appear to us to have been prepared for future conditions as if they predicted those conditions, but most readers will probably agree that individual neurons do not make nor represent predictions.

### 5.1. Background: Reservoir computing models

Our model is a simplified form of a reservoir computing model (Lukoševičius et al., 2012; Szary et al., 2011; Kello, 2013). The canonical reservoir computing network consists of three layers: an input layer, a reservoir layer, and an output layer. The reservoir layer typically contains a large number of nodes that are interconnected via non-updating random weights, with each node possessing a nonlinear activation function. We can understand the reservoir as a variant on Elman's (1990) SRN discussed in Section 4, where the context layer and the hidden layer are now one highly interconnected set of nodes, and no learning is performed on their weights. When a reservoir network is set up with a proper balance between excitation and inhibition (a "critical branching ratio"; Kello, 2013), spikes will propagate through the network without resulting in the network "freezing up" (i.e. all nodes becoming fully active or fully inactive). The resulting fluctuations of the network allow it to contain a kind of implicit, contextual memory, because spike patterns from the previous step will influence processing of the current inputs. Information about the present input as well as about past inputs is preserved in the intermixed, complex, instantaneous spike pattern of the reservoir.

Because spikes tend to propagate through these networks, when input is suddenly cut off from the reservoir layer, the endogenous activity of the network may persist for a few timesteps. This behavior is known as "fading memory." But critically, this kind of "memory," is not best thought of as a representation of the past. Instead, it is a kind of momentum from the past into the future, and therefore functions as a type of pattern completion in time. Under the right contexts, we suggest that such pattern completion may be indistinguishable from an ostensible "prediction."

The activity of the reservoir layer can be described as a projection of a relatively low-dimensional input pattern into a much higher-dimensional space. This high-dimensional mapping has the potential to carry a lot of information that can subserve complex mappings between inputs and outputs. However, the representation in the reservoir layer is complex, noisy, and distributed across many nodes, making it difficult to interpret directly. Therefore, canonical reservoir computing models must learn a mapping from the reservoir layer to a set of output nodes by virtue of a teaching signal. When this training signal is the next input pattern, we could fairly say that such a network is doing predictive processing (i.e. explicit use of predictions). More often, though, reservoir computing models are used to perform complex pattern classification functions, which are not thought of as predictive.

### 5.2. Model description

**Conceptual Overview.** There are three main ways in which our simple model deviates from the canonical reservoir computing model. First, it contains only two layers: an input layer and a reservoir layer. Using a small number of nodes, the activity patterns in our reservoir becomes easy to visualize and interpret, obviating the need to train an output mapping. Second, having no output layer means there is no training signal—the model is entirely unsupervised. Third, whereas canonical reservoir weights do not update, in our model, individual nodes continually adjust their own target activity level and the weights of connections with their neighbors in pursuit of homeostasis.

Conceptually, one can think of our nodes as individual agents that must obtain a minimum amount of energetic resources in order to survive. Energetic resources flow into the network via external perturbations. However, external perturbations do not reach every node, and even nodes that do receive external perturbations will not receive one on *every* time step. Furthermore, nodes will continually leak energy at a fixed rate. To deal with this variability in energetic influx, nodes that do not receive the minimum amount of energy from external perturbations will try to seek out input from neighboring nodes. This is represented in the model as an increase in positive connection weights with neighboring nodes.

However, too great an influx of energy to an individual is also undesirable. Nodes that receive *too much* energy have two strategies available simultaneously: they may try to decrease their input from neighbors (or even receive inhibitory input), and they may try to increase their "metabolism," dissipating more energy per unit of time. The latter strategy is implemented in the model as an increase in the "target" input value at each time step, and represents the idea that nodes can adapt to handle increased amounts of input. However, the target value has a floor, such that all nodes need *some* input to survive. Nodes will try to stabilize their actual input near their target value through some combination of input received from external perturbations, and input received from neighbors.

Critically, in order for nodes to stabilize their activity, they must receive the right amount of input at the right *time*. For example, a given node may be stimulated by an external perturbation that comes at  $t_0$  and  $t_3$ , but receive no external perturbation at  $t_1$  and  $t_4$ . Thus, at  $t_0$  and  $t_3$ , the node may need *inhibitory* input from neighbors to remain near its target activity level, while at  $t_1$  and  $t_4$  it may need *excitatory* input from neighbors. This means that nodes must learn to implicitly anticipate the temporal structure of perturbations, as well as the temporal structure of spiking activity in neighboring nodes, in order to stabilize their own activity.

**Network and Node Properties.** Our network contains a set of 100 processing nodes in the reservoir. Each node was randomly connected to other nodes in the reservoir with a link probability of  $P_{link} = 0.1$ , such that each node had approximately 10 neighbors. The weight of each link was randomly initialized by drawing from a normal distribution with mean 0 and s.d. of 1.

Each node  $n$  is characterized by 4 variables: a current activation level  $x_n$ , initialized at 0; (2) a fixed leak rate  $lr$  of 0.75 (e.g. if the activation level of a node is 1 at time  $t$ , the activation level will be 0.75 at time  $t + 1$ , in the absence of further input); (3) a variable target activation level, initialized at  $T_n = 1$ ; (4) and a variable spiking threshold  $T'_n$ , which was always equal to  $2T_n$  (e.g. when the target  $T_n$  was 1, the spiking threshold  $T'_n$  was 2). The value of the target  $T_n$  was given a lower bound of 1 (the value at initialization), ensuring that all nodes needed at least some continuous, positive input in order to remain near their target value.

When the activation level of a node is greater than or equal to the threshold value  $T'_n$ , it spikes, broadcasting a signal value of 1 to its neighbors. If the node does not achieve its threshold value, it fails to spike, broadcasting a signal of 0 to its neighbors. Thus, the network-endogenous input (i.e. excluding external perturbations to the network, discussed below) of each node  $n$  at time  $t$  was simply the sum of

$n$ 's weights with neighbors that spiked at time  $t-1$  (see Eq. (1) below). Activation cannot go below 0.

When a node spikes, it “dissipates” some activation by subtracting the threshold value. For example, if a node spikes with an activation level  $x_n = 10$ , a target value  $T_n = 5$ , and a threshold value  $T'_n = 10$ , its activation will immediately drop to 0. If the same node spikes with an activation level of  $x_n = 15$ , its activation will instead drop to 5. As we will describe further below, the effect of this rule is that, if a node is to remain close to the target activation level, its activity must either remain *under* the threshold value, so as to minimize the dissipation of activity, or go substantially *over* the threshold value, so as to cancel out the effect of dissipating activity upon spiking. Thus, neurons that spike with only a small margin will recruit more input from active neighbors, resulting in a pseudo-Hebbian effect whereby nodes increase weights with neighbors that fired on the previous time step.

**Input.** The input to this network was a sequence of “sentences” of the form [subject, verb, object, space]. There were two possible noun inputs (“man”, “dog”) that could serve as both subject and object, and two possible verb inputs (“walks”, “bites”). With the addition of an input encoding a “space,” this created 5 total inputs. The input sequences were generated by moving probabilistically through a transition matrix, shown in Fig. 7. This created 8 possible sequences that appeared with the probabilities shown in Table 1. These inputs were represented as 1-hot encodings across five input nodes (e.g. input ‘man’ is represented as [1, 0, 0, 0, 0]). Input nodes were connected to the main network through an input connectivity matrix, which was generated stochastically. Each input node was randomly connected to nodes in the reservoir with a link probability of  $P_{link} = 0.1$ . In addition to the randomly initialized internal weight matrix, these were the only sources of stochasticity in the model, which otherwise operates entirely deterministically subsequent to initialization.

Active input weights are set to a value of +5, and inactive weights are set to 0. This means that nodes received an external perturbation of magnitude +5 if they were connected to the current input pattern, and no external perturbation if they were not connected to the input pattern. Thus, each input node, when active (only one was active at a time), produced a perturbation of +5 to approximately 10 randomly selected reservoir nodes, and a perturbation of 0 to all other reservoir nodes.

**Activation Dynamics.** On each iteration, the input from external perturbation as well as from within the network is summed for each node. The activation vector  $x$  of the reservoir at time step  $t$  given input vector  $i$  can be written as follows:

$$x_t = x_{t-1} \cdot l + i \cdot W_i + s_{t-1} \cdot W_x \quad (1)$$

In Eq. (1),  $l$  represents the leak rate (0.75),  $W_i$  represents the input weight matrix,  $s_{t-1}$  represents the vector of spikes at the last time step (a 1 for a node that spiked, 0 for a node that did not spike), and  $W_x$  represents the recurrent weight matrix inside the reservoir. Importantly,  $W_x$  contains no self-connections, such that inputs come exclusively from network-external perturbations or from the spikes of neighbors. The

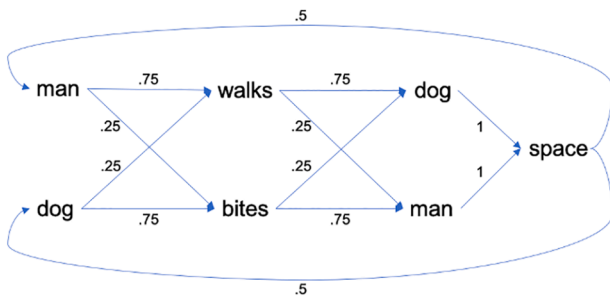


Fig. 7. Transition matrix used to probabilistically generate input sequences to the network, of the form [subject, verb, object, space]. Numbers adjacent to arrows indicate the probability of that particular transition.

Table 1

The set of 8 possible sentences that could be generated by the transition matrix and the probability of occurrence. Note that each sequence was always followed by a “space” input.

| Sequence      | Probability |
|---------------|-------------|
| Man walks dog | 0.28125     |
| Man walks man | 0.09375     |
| Man bites dog | 0.03125     |
| Man bites man | 0.09375     |
| Dog walks dog | 0.09375     |
| Dog walks man | 0.03125     |
| Dog bites dog | 0.09275     |
| Dog bites man | 0.28125     |

activation vector  $x$  is then compared with the threshold vector  $T'$  to compute the next spike vector  $s$ . Nodes that are above the threshold activation value result in a spike value of 1, and nodes that are below threshold result in a spike value of 0:

$$s_n = \begin{cases} 1, & x_n \geq T'_n \\ 0, & x_n < T'_n \end{cases} \quad (2)$$

For any neuron that spikes, the activation level will be decreased by the respective threshold level, prior to computing errors:

$$x(s) = x(s) - T'(s) \quad (3)$$

In Eq. (3),  $x(s)$  is the activation vector of neurons that spiked on the current iteration, and  $T'(s)$  is the corresponding threshold vector.

**Learning.** In the early time steps of our model, some reservoir nodes may receive one or more external perturbations on a given loop through our grammar, but many of the nodes in the reservoir will receive no external input. As such, input-connected nodes will initially reach their threshold activation level and therefore spike, but most nodes will undershoot both their target value and threshold value, and therefore will not spike. Based on the initially random internal weight matrix, activation will then spread throughout the network, continually intermixing with the activation from external perturbations. Since threshold values are initially relatively small ( $T' = 2$  for all nodes at initialization) relative to the magnitude of activation from input nodes (+5), nodes that spike may remain over their threshold value for subsequent iterations, despite dissipating some activation through spiking and some through leakage. This means that some nodes may spike repeatedly as a result of a single input, in the absence of recurrent negative feedback (see [Supplementary material](#)).

On each iteration, every processing node in the reservoir has an opportunity to adjust its weights with other processing nodes as well as its target activation value, with the homeostatic goal of reducing the discrepancy between actual and target values. (Input weights are never altered.) Using terms from control theory, what we are calling homeostasis amounts to a form of “automatic gain control,” whereby a processing unit attempts to maintain a stable output level despite variations in input.

The update for weights in the reservoir matrix  $W_x$  is determined by three things: the total error  $E$  (the difference between the actual and target activation; Eq. (3)), the number of weights available for updating  $N_n$ , and a learning rate  $L_{W_x}$ . Critically, nodes can only update connections with neighbors that spiked on the previous iteration. This means that if no neighbors spiked on the previous iteration, the number of weights  $N_n$  that can be updated is 0. Otherwise,  $N_n$  is equal to the number of neighbors that spiked on the previous iteration. The update for target value  $T$  (and, by extension, threshold values  $T' = 2T$ ) is determined only by the total error  $E$  and a learning rate  $L_T$ . Based on qualitative examination of piloting results, we set  $L_{W_x} = 0.1$  and  $L_T = 0.01$ , as these values allowed the model to stabilize around small error levels (mean  $E \sim 0.1$ ) relatively quickly (within 1000 sentences, or 4000 iterations) while avoiding the problem of overfitting (i.e. if learning

rates are too high, weights/targets may change drastically at every time step, making the model unable to reach an equilibrium).

$$E_{n,t} = x_{n,t} - T_{n,t} \quad (4)$$

Incoming weights to a node  $n$  from an active neighbor  $n'$   $W_{n,n'}$ ,  $n$  are adjusted so as to approach the target activation level according to the following learning rule:

$$W[n, n']_{t+1} = W[n, n']_t - \frac{E(n)_t L_{w_x}}{N_{n,t}} \quad (5)$$

Looking at the final term of Eq. (5), we can see that the total error  $E_n$  is distributed evenly across all  $N_n$  weights that are available for updating on a given iteration. See Fig. 8. When the error  $E_n$  is positive (a node overshoots its target), weights with active neighbors are reduced, and vice versa when error is negative. The inverse operation is applied to target, such that the target is *increased* when error is positive, and *decreased* when error is negative, according to the following equation:

$$T_{n,t+1} = T_{n,t} + E_{n,t} * L_T \quad (6)$$

Note again that  $E_n$  is computed *after* any node that has spiked on the current iteration has “dissipated” its threshold activation level  $T_n$ . For example, if a node spikes with an activation level  $x_n = 1.5 T$ , its activation after subtracting the threshold value will be exactly equal to its target value ( $1.5 T - T = .5 T = T$ ), and thus its error on the current iteration will be 0 ( $E_n = x_n - T_n = 0$ ), resulting in no change of the node’s parameters. We reiterate this point to emphasize that spiking does not necessarily correspond to a failure to maintain homeostasis for our nodes nor a “prediction error” (although some have proposed such an understanding of spiking activity, e.g. Fiorillo et al., 2014), but can also be a way for nodes to maintain homeostasis in the face of implicitly “expected” over-stimulation.

Finally, after parameter updates on each iteration, the next input is probabilistically selected based on the transition matrix encoding our “grammar,” and the entire process is repeated.

### 5.3. Outcomes

Because of the stochastic assignment of input patterns, separate runs of our model can produce distinct activity patterns. For present purposes, we are interested primarily in the “fading memory” property of this model, which can be seen extended for 3 or more iterations on the majority of runs. Interested readers may run our code themselves in order to observe other possible outcomes<sup>1</sup>.

However, to clearly illustrate the properties of this model, it will be helpful to first analyze the outcome of a single, representative run of the model in detail. Fig. 9 shows in red/yellow columns the spike pattern on the last 3 of 1000 training sentences of four inputs each ([*subject*, *verb*, *object*, *space*]) on one run (i.e. there were 4000 total training iterations). Then, the model was run for an additional 4 timesteps of testing (blue/yellow columns of Fig. 9), in which external input was provided for only the first timestep (which was a subject input, in this case ‘man’), and the model propagated its own activity for the final 3 timesteps with no further input (columns labeled ‘NA’ in Fig. 9).

If we visually inspect any two columns of Fig. 9 in which the same input was given, we can notice similar spike patterns recurring over time, which were only weakly present in the early iterations of the model, if at all (Supplementary Figs. 1 and 2). This can be seen even more clearly in the autocorrelation matrix from the same spike pattern over time (Fig. 10). In Fig. 10, we can see that the highly correlated spike patterns occur in response to different instances of the same input in the same position of a sentence, which could be read as emergent population codes for each input. We found that these population codes were quite

consistent over at least the final 100 timesteps of the model (autocorrelations were between 0.65 and 0.77; see Supplementary material).

The crucial behavior, for our purposes, occurs when the external input is turned off in the final three iterations, seen in the blue-and-yellow columns of Fig. 9, and in the ‘NA’ rows of Fig. 10 (demarcated by dotted lines at the bottom). Despite having received no input, the network itself generates a spike pattern that resembles a slightly-degraded version of the population codes corresponding to the inputs that *could* have appeared in those positions. For example, the first iteration without input on this run, based on the transition matrix, would most likely have been “walks” with  $P = .75$ , or “bites” with  $P = .25$ . The pattern that emerges in the absence of input (the second blue column in Fig. 9 and the first NA row in Fig. 10) is highly correlated with the previous instance on which “walks” was actually presented ( $r = 0.60$ – $0.66$ ), and the next most highly correlated pattern coincided with previous instances where the verb was “bites” ( $r = 0.34$ ). This effect continues, though fading slightly, for the following 2 test iterations, which would have most likely been “dog” followed by a “space” input. In the absence of input, the network generates a pattern that highly resembles the most-likely completion of the terminated input sequence, and slightly resembles other less-likely sequences.

### 5.4. Fading memory “Predictions”

To evaluate the consistency of this effect, we conducted 500 independent runs of the model. At the end of each run, we saved the final state of the model, then conducted 6 tests during which one or two subsequent iterations of input were provided (e.g. [‘man’, no input] or [‘man’, ‘walks’, no input]). We then computed the average correlation between the spike pattern that appeared after input was cut off, with the previous spike patterns that coincided with each input/position combination. If population codes emerge in this homeostatic reservoir network, and if the network does pattern-completion through time, we would expect the spike patterns on the final timesteps where *no* input was given to nonetheless resemble previous spike patterns corresponding to the inputs that could have appeared in those positions. For this analysis, we considered only the last 100 training sentences (400 iterations), to account for the fact that, while there appear to be emergent population codes that are relatively stable for a period, these may continue to change over longer time spans. The results presented in Table 2 represent the grand average of these correlations over all 500 runs.

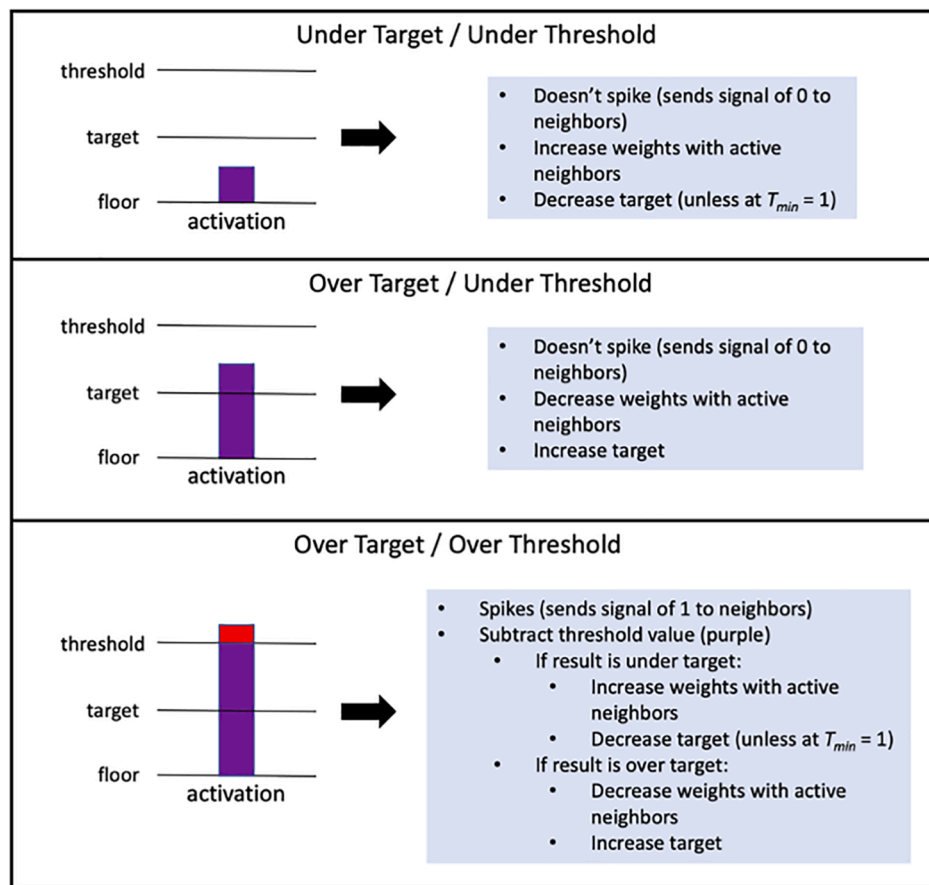
As Table 2 shows, the “fading memory” output of our model tends to resemble the population codes for the input that would have most likely appeared in the next position, if the input had not been cut off. For example, given input of ‘man’ followed by no input, the activity of the network is most highly correlated with previous instances where ‘walks’ appeared (which followed ‘man’ 75% of the time). The next most similar population code was ‘bites’, which appeared 25% of the time following ‘man’ during training.

Interestingly, we can see that the model’s fading memory output tends to be specific to the position at which the input is received. For example, given [‘man’, ‘walks’], the output is most highly correlated with instances where ‘dog’ appeared in the object position (and, slightly less so, with instances where ‘man’ appeared in the object position), but has a much lower correlation with the activity observed for instances where ‘dog’ appeared in the *subject* position. Importantly, the external perturbation was the same in these two cases. This again points to the fact that our model became sensitive to the temporal structure in inputs, rather than simply the input patterns themselves.

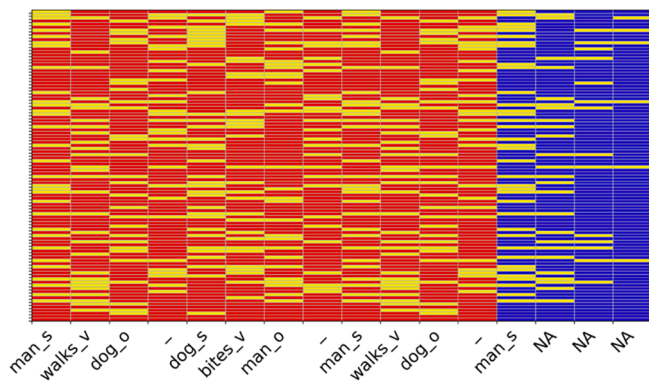
Furthermore, we can notice that the fading memory output appears sensitive to the overall frequency with which different sentences could occur. For example, given [‘man, walks’] or [‘dog, walks’], an input of ‘dog’ appeared in the next position 50% of the time. However, the fading memory patterns are slightly different in these two cases: the output is more highly correlated with previous instances of ‘dog<sub>obj</sub>’ when the

<sup>1</sup> <https://github.com/bfalandays/HomeostasisModel>





**Fig. 8.** A schematic showing the rules for weight/target updating, depending upon a node's current activation level relative to the node's current target level and spiking threshold value (which was always twice the current target level).



**Fig. 9.** The spike pattern for the last 3 sentences (12 iterations) of training (red/yellow columns) on a representative run of the model, plus four test iterations of testing during which external input was cut off after the subject noun (blue/yellow columns). Nodes are arranged on the y-axis, and time is on the x-axis. Active nodes are shown in yellow, with inactive nodes in red/blue. For the last three iterations, 'NA' is shown in the column labels to indicate that no input was provided. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

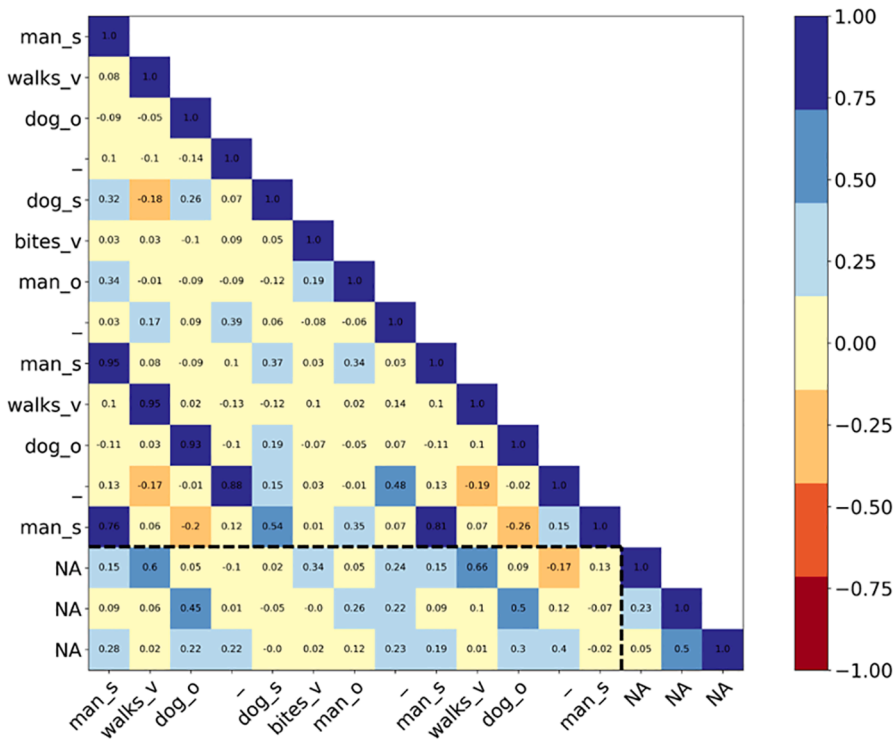
network was probed with ['man, walks'] ( $r = 0.452$ ) than when it was probed with ['dog, walks'] ( $r = 0.402$ ). In this way, the network appears sensitive to the fact that the overall probability of observing the sequence ['man', 'walks', 'dog'] ( $P = .28$ ) is higher than the probability of observing ['dog', 'walks', 'dog'] ( $P = .09$ ).

This pattern completion behavior emerges because the model comes

to depend upon its own self-generated activity to keep nodes poised near their target activation value. In fact, the external perturbations ultimately account for very little of the total activation pattern at each time step, most of which is driven endogenously. As such, when the external input shuts off, many of the nodes that constitute the population code for what would have been the *next* external input will derive their incoming activation mostly or entirely from the *previous* spike pattern. Because there is repeated structure in time, learning to stabilize present states based on the past is functionally equivalent to predicting the future. In other words, by adapting to past inputs, our nodes are learning what to do *next time* that pattern occurs in order to remain close to their target activation value. Needless to say, adaptation would not look like a form of prediction if the past conditions that spurred adaptation did not recur in the future. Thus, our model is not so much *predicting* external input as it is *entraining* to the temporal structure of external input.

### 5.5. Hallmarks of predictive coding

In addition to generating prediction-like effects in its fading memory, our model also exhibits a signature of predictive coding that Luthra et al. (2021a) and Luthra et al. (2021b) also saw in TRACE: decreased activation for predictable inputs relative to unlikely inputs. We took the average activation of nodes in our network as a reasonable analog to an EEG signal from a real brain. We tested for a signature of predictive coding in two ways. First, Fig. 11 shows the mean activation of the network, obtained from the final 400 timesteps (100 sentences) of training for 500 distinct runs of the model, as a function of the transitional probability of the sequence (x-axis) and the item type (subject, verb, object, or space; colors in Fig. 11). This plot shows that mean activations were higher when low-probability transitions were observed.



**Fig. 10.** The autocorrelation matrix for the last 12 iterations of training, plus four test iterations during which only a subject noun was provided as input. Before the input turns off (above/left of the dotted lines), we can see that the reservoir has developed population codes—highly correlated spike patterns for separate instances of the same input. When the input is turned off (below/right of the dotted lines), we can see that the fading memory produces spike patterns that remain highly correlated with the likely next inputs, as if the network generates the pattern it “expects.”

**Table 2**

Fading memory “predictions” at the end of 1000 training loops (4000 iterations). When input is cut off after one or two iterations, the model’s fading memory produces spike patterns that are most highly correlated (bold cells) with the spike patterns observed for previous instances of the input that would be most likely to appear in the next position (e.g. input of ‘man’ followed by no input results in a pattern resembling previous instances in which ‘walks’ was actually presented). The next most-highly-correlated input pattern corresponds to the next-most-likely pattern that would have appeared (e.g., the fading memory after input of ‘man’ is also correlated with ‘bites’).

| Test Input Presented | Possible Inputs at Each position [subject, verb, object, space] |                       |                         |                         |                      |                      |       |
|----------------------|-----------------------------------------------------------------|-----------------------|-------------------------|-------------------------|----------------------|----------------------|-------|
|                      | ‘man’ <sub>subj</sub>                                           | ‘dog’ <sub>subj</sub> | ‘walks’ <sub>verb</sub> | ‘bites’ <sub>verb</sub> | ‘dog’ <sub>obj</sub> | ‘man’ <sub>obj</sub> | Space |
| [man]                | 0.099                                                           | 0.016                 | <b>0.467</b>            | 0.319                   | −0.009               | 0.065                | 0.173 |
| [man, walks]         | 0.06                                                            | 0.061                 | 0.17                    | 0.0004                  | <b>0.452</b>         | 0.213                | 0.036 |
| [man, bites]         | 0.049                                                           | 0.052                 | −0.008                  | 0.199                   | 0.237                | <b>0.402</b>         | 0.038 |
| [dog]                | 0.02                                                            | 0.107                 | 0.3173                  | <b>0.469</b>            | 0.069                | −0.01                | 0.176 |
| [dog, walks]         | 0.057                                                           | 0.0546                | 0.205                   | −0.003                  | <b>0.402</b>         | 0.237                | 0.04  |
| [dog, bites]         | 0.06                                                            | 0.06                  | 0.002                   | 0.171                   | 0.213                | <b>0.452</b>         | 0.039 |

Avg. Correlation Coefficient b/w “Fading Memory” Output and Prior Instances of Each Input-at-a-position (End of Training, 500 runs).

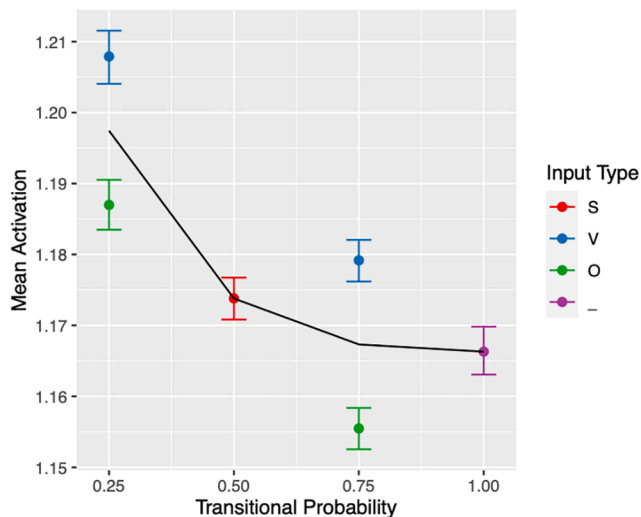
We can also see that activations were higher for verb inputs, relative to object noun inputs. This may be due to the fact that the Shannon entropy of subject nouns was 1 bit (each subject noun could appear with  $P = .5$ ), while the entropy of verbs and object nouns was 0.81 bits (each appearing with  $P = .25$  or  $0.75$ ). This means that verbs were preceded by more surprising inputs, in an information theoretic sense, than were object nouns. If activation is higher for surprising inputs, and this activation may extend into the next timestep, then it is reasonable to expect activations to be lower, on average, at the object position of a sentence compared to the verb position.

We next considered what may happen when the network is presented with an “ungrammatical” sequence—a sequence of items that was never encountered in training. During training, the network always observed sequences of the form [subject, verb, object, space]. We conducted a test using these same elements in a new order, [subject, object, verb, space]. In this new order, the first and fourth items are consistent with the sequences observed in training, while the second and third inputs were never observed in those positions during training. In Fig. 12, we have plotted the average activation of the network during the final 100 timesteps of training (25 sentences of four elements each; to left of the

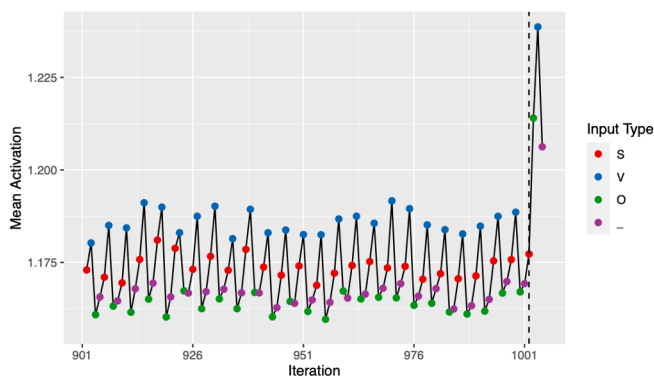
dashed line), followed by the average activation of the network given each of 8 possible ungrammatical sequences. These values were averaged over 500 distinct runs of the model. The mean activation of the network for the final 100 timesteps of training was 1.17 (SD = 0.009). As Fig. 12 reveals, activation increases substantially in response to the first ungrammatical input, increases further with the second ungrammatical input, and drops at the conclusion of the sequence with a space input, which was consistent with the patterns observed in training. The mean activation values in response to the ungrammatical inputs are greater than 1.2, more than 3 standard deviations above the average activation in the final timesteps of training. This shows that the network’s response to input sequences with a cloze probability of 0 are similar to the ones observed with low cloze probability (e.g., when transitional probabilities were 0.25 in Fig. 11).

## 5.6. Discussion

The simplified reservoir computing model presented here has no externally-imposed target activation pattern (in the form of a teaching signal), and yet consistent distributed “population codes” naturally



**Fig. 11.** The average activation value of nodes as a function of the transitional probability of the observed input pattern. Colors correspond to possible sentence positions (subject, verb, object, or space). Each subject input appeared with 0.5 probability, and a space (“\_”) input always appeared in the fourth position of a sequence. Object and verb inputs had associated transitional probabilities of either 0.25 or 0.75. The black line shows the mean value, averaging across input types. Error bars represent a 95% confidence interval around means obtained from the final 400 timesteps of training across 500 distinct runs of the model.



**Fig. 12.** The average activation value of nodes as a function of iteration, for the final 100 iterations of training (left of the dashed line), plus a final sequence (four iterations) of testing with a sequence order that never appeared in training (subject, object, verb, space). Colors correspond to the item type. Points to the left of the dashed line represent averages over 500 distinct runs of the model. Points to the right of the dashed line represent averages over 8 possible ungrammatical sequences presented to 500 distinct runs.

emerge. Through a learning algorithm by which individual nodes pursue a constant level of activity near an arbitrary target, the population code present at any given time point becomes critical for generating the full population code at the *next* time step, only a small subset of which may be partially attributable to external perturbation. As such, the “fading memory” of our reservoir tends to continue the established sequential pattern for a few time steps in the absence of any input. We also observe one of the ostensible signatures of predictive coding—a decrease in signal when input sequences are predictable and an increase in signal when the input is unexpected—despite the fact that our network does not explicitly generate predictions.

While predictive coding models of neural computation take neural activity (largely, spiking behavior) to encode prediction errors (e.g., FIORILLO et al., 2014), our learning algorithm treats spiking as a viable way to *maintain* homeostasis, so spikes in our model do not necessarily

encode errors. Nonetheless, we find the ostensible signature of predictive coding in our model, with greater activity in the network in response to lower-probability or 0-probability (ungrammatical) sequences. Given that this pattern can emerge in systems that do not make use of predictions nor prediction errors, such as TRACE and our reservoir model, and given that it may sometimes *not* emerge in systems that are explicitly trained to predict, such as an SRN (Luthra et al., 2021a; Luthra et al., 2021b), caution may be warranted before interpreting similar patterns in human data as support for a “realist” stance on predictive coding in the brain (see Colombo and Seriès, 2012).

If we didn’t know any better, it would be tempting to say that this network has learned to “predict” upcoming inputs. However, while we can describe this form of adaptation as *functionally* predictive, it is important to note that what the system is doing is simply pattern completion, where the network learns to generate patterns that contribute to local homeostasis in the midst of variable external perturbations. As such, the activity of the network is not literally an encoding of predictions or prediction errors. Nor is the network simply encoding inputs, since much of the activity is endogenously driven. Rather than exhibiting “weak anticipation” by generating predictions of future input that are labeled as such, this homeostatic reservoir network does “strong anticipation” (Dubois, 2003; Stepp and Turvey, 2010) by coupling with its environment instead of representationally modeling it. This illustrates the fact that systems can be implicitly anticipatory, without forming explicit predictions.

## 6. Predictive processing emerges from multi-scale pattern completion

Predictive processing and predictive coding both appear to be crucial elements of learning and cognition (Elman, 1990; Rao and Ballard, 1999). However, just because a cognitive faculty appears to be a singular coherent ability that stands out amid other abilities does not mean that the brain has devoted a singular coherent mechanism dedicated solely to that ability (Uttal, 2001; Gibbs and Van Orden, 2010). To some people, master chess playing is a singular coherent ability that stands out amid other abilities, but few would propose that there is a chess-playing module in the human brain. Brain evolution conserves mechanisms over millennia (Anderson, 2014), such that if a pre-existing capacity (such as pattern completion among connected neurons) can serve the purpose of predictive processing, then evolving an additional separate mechanism dedicated to predictive processing may be unnecessary. Perhaps certain aspects of the predictive processing that appear to be so ubiquitous in language processing (Onnis et al., 2008), motor movement (Wolpert and Flanagan, 2001), and cognition in general (Clark, 2015; Friston, 2019) are actually due to multi-scale pattern completion of the future, rather than due to an inference mechanism that is designed to generate representations of predicted input and then calculate an error signal. That said, it seems clear that in certain circumstances the mental representation of an anticipated future event can become sufficiently overt and explicit that the cognizer is consciously aware of it as a prediction (Brothers et al., 2017; Huettig, 2015). Under these longer time scale circumstances, the pattern completion may spread so widely so as to not only involve sensory features of not-yet-presented stimuli but to also pre-activate conceptual and linguistic features associated with those stimuli as well.

Instead of treating pattern completion as an alternative account in direct conflict with the “Bayesian brain,” we prefer to treat pattern completion as the mechanistic neural instantiation of the computational principle of probabilistic prediction. Importantly, when one looks at this particular neural instantiation of that powerful computational principle, it appears much less representational, or deliberative, or intelligent than the “Bayesian brain” framework might lead one to infer. Just about any neural net can do pattern completion – including those inside an octopus or a worm. Thus, when one uses a Bayesian predictive model to describe a cognitive or sensorimotor process, it might be worth taking into

account the fact that the “predictions” it generates need not to be treated as hypothesized representations generated by a cognitive entity that knows they are hypotheses. The system doing the predicting may be exhibiting “strong anticipation” (Dubois, 2003), which does not require any modeling or representing of the pattern being anticipated. Those parallel graded predictions might be little more than “pattern completion in the temporal domain.”

In this article, we have discussed a variety of examples wherein predictive processing appears to emerge from systems that are merely adapting themselves to their environment and utilizing the pattern completion capabilities availed by their interconnected sub-units. Whether they are loosely-coupled oscillators or loosely-connected nodes in a network, these systems clearly do not have a mechanism dedicated to making predictions. And yet they exhibit predictive processing, anticipating future events. For example, Elman and McRae (2019) extensively trained a connectionist model of event knowledge, using recurrent connections in its hidden layer (and feedback connections to the input layer) to fill in the gaps in the event sequences to which it was exposed. With events such as John Cuts Himself, the pattern completion capabilities of this network allowed it to partially activate appropriate (but unmentioned) Instrument nodes, such as *ax* and *knife*, as well as future Action nodes, such as *bleeds* and *dies*. Thus, without a mechanism dedicated to inference-making, it was nonetheless capable of making the equivalent of elaborative inferences by activating nodes that correspond to event elements (such as inferred Instruments, and future Actions) that were not yet present in the stimulus. In this way, their network “implemented pattern completion in the moment, and pattern completion through time (i.e., prediction)” (McRae et al., 2021).

In two sample simulations with the TRACE model of spoken word recognition, we demonstrated that the pattern completion capabilities of that network were able to activate the final phonemes of a spoken word that ended halfway through its delivery. When the network received only “æda” as input, it naturally spread some partial activation to the phoneme nodes for /p/ and /t/ as part of the word “adapt” (Fig. 4). When the network received only “prag” as input, it naturally spread some partial activation to the phoneme nodes for /r/ and /ə/ and /s/ as part of the word “progress” (Fig. 5). Thus, without any mechanism dedicated to prediction, the TRACE network nonetheless completes the pattern of a word that was only partially presented, by partially activating phonemes that it should be receiving in the near future.

In a second existence proof of how a system can exhibit “prediction for free,” we introduced a simplified reservoir network of nodes with the sole mission of maintaining homeostasis (a form of automatic gain control). After training on sequences of inputs generated from a simple, probabilistic grammar, this network learns to balance its internal propagation of activity with external perturbations, and anticipation of future input naturally emerges. After training, when the network receives only a partial sequence as input, the internal activity of the network nonetheless produces a pattern that is quite similar to the one we would see if the network *had* received an appropriate input – as if the network “knew” what was coming next. The network also exhibits an increase in activity in response to lower-probability sequences, and sequences that were never presented in training, which is often taken as a signature pattern of predictive coding. Importantly, neither prediction nor memory were explicit features of this network. The only mechanism driving this emergent anticipation is a few basic rules for maintaining homeostasis.

We situate this network and its free, inherent predictive processing amongst a suite of anticipatory systems and behaviors (Elman and McRae, 2019; Tang et al., 2018; Spivey, 2008; Stepp and Turvey, 2010; Stephen et al., 2008; McClelland and Elman, 1986; Magnuson, 2015). We pay particular attention to language processing to provide an alternative to traditional accounts in this domain, however, we maintain that the predictiveness of pattern completion is generalizable across a range of cognitive and perceptual processes. Furthermore, the pattern completion paradigm is easily applicable to many spatiotemporal scales

of analysis, whether at the scale of phonemes or words or sentences, or at the scale of visual features or objects or scenes. Our account is complementary to others that have described prediction as attuned to and emerging from context-sensitivity (Kirchhoff and Robertson, 2018; Dubois, 2003; see also: Ferreira & Qiu, submitted; Huettig et al., submitted). The notion of multi-scale pattern completion extends our understanding of prediction by showing that prediction is understanding, extended through time.

## Acknowledgments

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## Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.brainres.2021.147578>.

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